

Q1

$$a = 4, b = 3/8, c = 2$$

Q2

$$x = [4 \pm \sqrt{16 + 12(2+\sqrt{5})(2-\sqrt{5})}] / [2(2+\sqrt{5})]. \text{ This gives } x = (4 \pm \sqrt{16-8}) / (2(2+\sqrt{5})) : (4 \pm \sqrt{8}) / (2(2+\sqrt{5}))$$

Q3a

$$\text{leading coefficient} = +3, \text{ one factor} = (x+2), \text{ remaining factors} = (x-2)(x+4)$$

Q3b

$$\text{Case 1: } 5x-2 \geq 4x+1 \text{ gives } x \geq 3 \text{ (critical value 3). Case 2: } 5x-2 \leq -4x-1 \text{ or. Combined solution: } x \leq 3$$

Q4a

$$r = 12 \text{ and } \frac{1}{2}r^2 = 57.6. \text{ Solving: } r = 8, \quad = 1.5$$

Q4b

$$AC = 20. \text{ Shaded area} = (\frac{1}{2} \times 20 \times 9.6) - 57.6 = 96 - 57.6 = 38.4$$